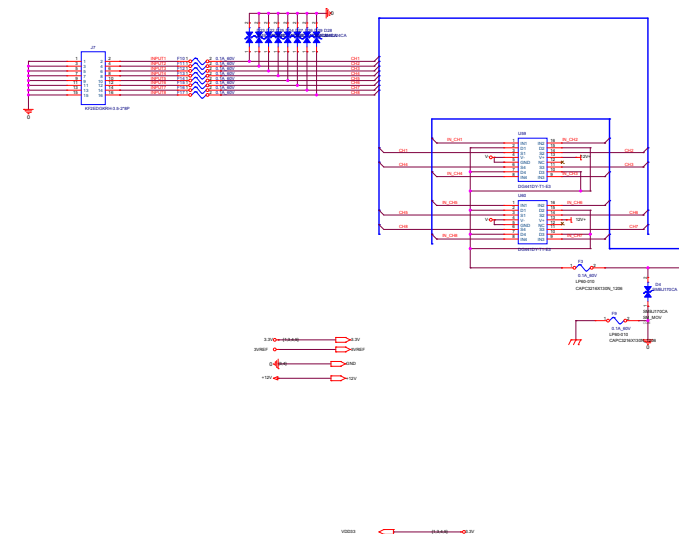


Inputs



Switch Table

Mode	SW1 (ACV TRMS)	SW2 (ADP)	SW3 (GPR)	SW4 (GVS)	SW5 (FTH)
DCV	ON	OFF	ON	OFF	ON
ACV	OFF	ON	OFF	ON	OFF
DCA	ON	ON	ON	ON	ON
ACA	OFF	OFF	OFF	OFF	OFF
Resistance	ON	ON	ON	ON	ON
Capacitance	ON	ON	ON	ON	ON
Frequency	ON	ON	ON	ON	ON
Capacitance	ON	ON	ON	ON	ON
ADP	ON	ON	ON	ON	ON

1/8 channel selection

Channel 1

Channel 2

Channel 3

Channel 4

Channel 5

Channel 6

Channel 7

Channel 8

Test mode selection

Switch 2

Switch 3

Switch 4

Switch 5

Measurement Mode Switching

The diagram illustrates the Measurement Mode Switching circuit. It consists of five channels (FC1 to FC5) and a RANGE channel, each using a TLP2309 optocoupler. The channels are connected to a common bus (PMS, PMS, PMS, PMS, PMS) which is connected to a ULN2803B 101 STR. The diagram also shows a detailed view of the TLP2309 internal structure and a table of measurement modes and functions.

Measurement Mode Switching

Measurement mode depends on the logic level of SLACDC, FC1, FC2, FC3, FC4, FC5, and KEY. When FC5 is high, the measurement mode list is below.

SLACDC	FC1	FC2	FC3	FC4	Mode	Function of KEY
0	1	0	1	1	Voltage Measurement	DCV → ACV
0	1	1	0	1	Auto DC Current Measurement(μA)	DCA → ACA
0	1	1	1	1	Auto DC Current Measurement(mA)	DCA → ACA
0	0	0	0	0	22A DC Current Measurement(A)	DCA → ACA
0	1	1	1	0	DCmV 220.0mV	DCmV → ACmV
0	1	1	0	0	Manual DC 220.000V	DCA → ACA
0	1	0	0	0	Manual DC 220.00A	DCA → ACA
0	1	0	0	1	Manual DC 2200.0A	DCA → ACA
0	0	0	0	1	Resistance Measurement	G → Continuity
0	0	0	0	1	Continuity Check	Continuity → Diode
0	0	1	1	1	Resistance Measurement	G → Continuity → Diode
0	0	0	1	0	Frequency Measurement	—
0	0	1	1	0	Capacitance Measurement	—
0	0	1	0	0	Temperature Measurement	°C → °F
0	0	1	0	1	Resistance Measurement	G → Diode
1	1	0	1	1	Voltage Measurement	ACV → DCV
1	1	1	0	1	Auto AC Current Measurement(μA)	ACmV → DCmV
1	1	1	1	1	Auto AC Current Measurement(mA)	ACmV → DCmV
1	0	0	0	0	22A AC Current Measurement(A)	ACA → DCA
1	1	1	1	0	ACmV 220.0mV	ACmV → DCmV
1	1	1	0	0	Manual AC 22.000V	ACA → DCA
1	1	0	0	0	Manual AC 220.00A	ACA → DCA
1	1	0	0	1	Manual AC 2200.0A	ACA → DCA
1	0	0	0	1	Manual AC 22000.0V	ACA → DCA
1	0	0	1	1	ADP9220.0V	—
1	0	0	1	1	ADP9220.0V	—
1	0	1	1	1	ADP9220.0V	—
1	0	1	0	0	ADP9220.0V	—
1	0	1	1	0	ADP9220.0V	—
1	0	1	0	0	Temperature Measurement	°F → °C
1	0	1	0	1	Capacitance Measurement (Clamp)	—

Measurement mode depends on the logic level of SLACDC, FC1, FC2, FC3, FC4, FC5, and KEY. When FC5 is high, the measurement mode list is below.

SLADC	FC1	FC2	FC3	FC4	Mode	Function of KEY
0	1	0	1	0	Voltage Measurement	DCV → ACN
0	1	1	0	1	Auto DC Current Measurement(A)	DCA → ACA
0	1	1	1	1	Auto DC Current Measurement(A)	DCA → ACN
0	0	1	0	0	22A DC Current Measurement(A)	DCA → ACA
0	0	1	0	0	DCmV 220.0mV	DCmV → ACmV
0	1	1	0	0	Manual DC 22.000A	DCA → ACA
0	1	1	0	0	Manual DC 220.00A	DCA → ACA
0	1	0	1	0	Manual DC 2200.0A	DCA → ACA
0	1	0	0	1	Manual DC 22000A	DCA → ACA
0	0	0	1	1	Resistance Measurement	G → Continuity
0	0	0	1	1	Continuity Check	Continuity → Diode
0	0	0	1	1	Resistance Measurement	G → Continuity → Diode
0	0	0	1	0	Frequency Measurement	
0	0	1	1	0	Capacitance Measurement	
0	0	1	0	0	Temperature Measurement	C → F
0	0	1	0	1	Resistance Measurement	G → Diode
1	1	0	1	1	Voltage Measurement	ACV → DCV
1	1	1	0	1	Auto AC Current Measurement(A)	ACIA → DIuA
1	1	1	1	1	Auto AC Current Measurement(A)	ACIA → DI mA
1	1	1	0	0	22A AC Current Measurement(A)	ACIA → DCA
1	1	1	0	0	ACmV 220.0mV	ACmV → DCmV
1	1	0	1	0	Manual AC 22.000A	ACA → DCA
1	1	0	0	0	Manual AC 220.00A	ACA → DCA
1	1	0	1	0	Manual AC 2200.0A	ACA → DCA
1	1	0	0	1	Manual AC 22000A	ACA → DCA
1	0	0	1	1	ADPQ(220.0)	-
1	0	0	0	1	ADP(1220.0)	-
1	0	1	1	1	ADP(320.0)	-
1	0	0	1	0	ADP(22.00)	-
1	0	1	0	0	ADP(2.000)	-
1	0	1	0	0	Temperature Measurement	F → C
1	0	1	0	1	Capacitance Measurement (CapnF)	

